

Do the gains from digital technology waves differ by income level?

A systematic review with a pilot meta-regression

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Draft status (remove before submission). All study data in this draft were extracted by machine from the full texts. Every printed number used in the meta-regression was checked against the text twice, and 64 of the 67 study records were re-read once by a separate model run, but *no human has verified any extraction*. Screening used one assisted reviewer and no second screener. The protocol was not registered before the search was run. Scopus and Web of Science have not been searched. Sections marked [author] need the author's input. Numbers in this draft come from the files in the project repository, and the repository is the reference if text and data disagree.

Abstract

Background. Mobile telephony, the internet and, more recently, artificial intelligence (AI) are often expected to widen or narrow income gaps between countries, but the empirical cross-country evidence has not been synthesised across outcomes. **Methods.** We ran a systematic review following PRISMA 2020, searching OpenAlex, Crossref and NBER for 1995 to 2026, and assessed 107 full texts. We extracted effects on six outcome families, rated risk of bias with an adapted ROBINS-I tool, synthesised results narratively, and ran a pilot meta-regression of partial correlations for ICT-type adoption and output or productivity. **Results.** Sixty-seven studies met the criteria (54 on ICT, mobile, internet and broadband; 11 on AI; 2 on both). None was rated at low risk of bias (moderate 8, serious 50, critical 9). Most studies report a positive association between ICT-type adoption and output, but the pooled partial correlation across 16 studies is 0.19 (95% CI 0.10 to 0.27) with I^2 of 93%, and falls to between 0.005 and 0.06 after correcting for small-study bias. Estimates from GMM are on average 0.16 lower than those from other estimators. Twenty-six studies compare income groups or regions: eight find larger effects in richer economies, six in poorer ones, and three find no difference, mostly without a formal test. Returns are conditional on human capital, financial depth, governance and infrastructure. Evidence on AI comes from 13 studies, mostly short and using proxies. In an exploratory World Bank panel for 1995 to 2019, mobile adoption was associated with faster growth in every income group, most strongly in lower-middle income economies, but the pattern did not survive a predetermined specification, and an event-study design was not interpretable. **Conclusions.** The evidence does not establish that rich or poor economies gain more, and it cannot support causal statements. Credible identification for low- and middle-income economies is the main gap.

Keywords: digital technology; ICT; artificial intelligence; income level; systematic review; meta-regression.

JEL codes: O33, O47, O11, L96.

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1 Introduction

Whether new technologies narrow or widen income gaps between countries is a long-standing question, and it has returned with artificial intelligence. Early empirical work found sizeable output effects of telecommunications infrastructure [1]. Later work on the long-run diffusion of technologies finds that technologies arrive at similar times in rich and poor countries, but that the intensity of their use diverges [2, 3]. If so, gains from a new technology should depend on a country’s conditions at the time of adoption, and not only on whether the technology has arrived.

Evidence on this is scattered. Studies use different technologies (mobile subscriptions, internet users, composite ICT indices, ICT capital, robots, AI readiness indices), different outcomes (growth, productivity, employment, inequality, human development, emissions), different samples and different estimators. Individual studies often report an average effect and one or two interaction terms. A reader who wants to know whether a low-income economy can expect the same payoff as a high-income one has no single place to look.

We therefore ask: *in empirical cross-country studies of digital and general-purpose technology waves, do effects on output, structural change, labour markets, distribution, living standards and resource use differ by income level and by country characteristics, how large are the differences, and how credible is the evidence?* We keep studies of historical waves (ICT, mobile, internet, broadband, crypto-assets) separate from studies of AI, because AI evidence is younger and thinner.

The contribution is a structured map of the evidence with a quality rating of each study, a transparent count of where income-group comparisons exist, and a pilot meta-regression of effect sizes for the best-covered question (ICT-type adoption and output). We are explicit about what the data cannot show. All data and code are in the project repository.

2 Methods

2.1 Protocol and registration

The review follows PRISMA 2020 [4] and the structure of PRISMA-P [5]. The protocol (version 1.0, 2 October 2026) is in the repository. It was intended to be registered on the Open Science Framework before the search, but registration had not been completed when the search was run. This is a deviation from good practice, and we log the amendments made after the search in Appendix B. **[author: register the protocol on OSF and state the registration number or explain]**

2.2 Eligibility

We included quantitative empirical studies with their own estimation (panel regression, instrumental variables, growth accounting, event studies, structural estimation) that relate adoption, diffusion, investment or capability of ICT in general, mobile telephony, internet or broadband, crypto-assets or AI to at least one outcome in six families: (1) output and productivity, (2) structural change, (3) labour market, (4) poverty and distribution, (5) living standards beyond income (consumption, health, schooling, financial inclusion) and (6) resource cost (electricity use, carbon intensity). Historical-wave studies needed at least 10 economies or an explicit income-group comparison; AI studies needed at least 3 economies or an income-level compari-

son. Studies from 1995 onward in English were eligible. Purely conceptual papers, reviews and studies whose only outcome was adoption itself were not synthesised.

2.3 Search and screening

We searched OpenAlex (a Boolean query restricted to the economics field), Crossref (96 technology-by-context queries) and the NBER working-paper series, covering 1995 to the search date of 2 October 2026. arXiv was attempted but did not respond in usable time. Scopus, Web of Science and EconLit were not searched. After removing duplicates, 3,612 records remained. Screening was done by one assisted reviewer: automated rules removed records with no eligible technology term, no outcome term or no multi-economy cue (stage 1), then title and abstract screening applied the eligibility table. No second screener was available, so we report no agreement statistic. A random verification sample of 90 decisions is in the repository awaiting an independent check.

Full texts were obtained from open-access sources, repositories and the author. Records without a free full text, and those the author judged to come from unreliable venues after visiting the journal site, were not read and are not used as evidence. This creates a selection mechanism that favours open-access studies, discussed in Section 5.

2.4 Data extraction and risk of bias

For each included study we extracted technology, exposure measure, sample, estimator, identification strategy, results by outcome family with printed estimates and uncertainty, heterogeneity findings, and group-specific results, using a fixed form. Extraction was done by a language-model agent working from converted full texts, with a locator and a short verbatim quote for every result. An automatic check matched 403 of 586 quotes verbatim across all 107 records read (the rest are affected by table layout and paraphrase). A second, separate model run then re-read each of 64 included records and each of 203 regression estimates against the text and classified every item as confirmed, corrected or unverifiable. Three studies added late (71, 278 and 341) were not given this second pass; we checked their key printed numbers by text search.

Risk of bias used a six-domain tool adapted from ROBINS-I [7]: confounding and reverse causation, exposure measurement, selection of countries and years, outcome measurement, specification and researcher degrees of freedom, and selective reporting. Overall ratings were low, moderate, serious or critical.

2.5 Synthesis

We synthesised results narratively, following the SWiM reporting guideline [6], by outcome family and by technology stream. Counts of study directions are descriptive; they are not weighted and are not effect sizes.

For ICT-type adoption and output or productivity we ran a pilot meta-regression in the tradition of Stanley and Doucouliagos [8]. Each estimate was converted to a partial correlation coefficient, $PCC = t/\sqrt{t^2 + df}$, with df equal to the number of observations minus 10 regressors (a sensitivity check uses 0, 5 and 20). The t -statistic came from the printed coefficient and standard error, or from a printed t -statistic. Estimates reported with p -values or intervals only were not converted. We excluded interaction terms, crypto and AI exposures, estimates without a usable number of observations, and five estimates judged not poolable on the second pass (a fintech-credit exposure, a growth outcome mixed with level outcomes, and a TFP index in levels). We pooled one headline estimate per paper by random effects [10], ran a weighted moderator

regression with standard errors clustered by paper, and tested for publication selection with FAT-PET and PEESE [9]. With 18 papers, the clustered standard errors are optimistic and p-values are indicative.

2.6 Additional exploratory questions

Beyond the main question we specified nine further questions and their analyses before running them (`review/extra_questions_spec.md`), all using the data already collected: whether the pooled association depends on the measurement of the technology (a composite index versus a single indicator), on the outcome form (growth versus level) and on the sample period (added one at a time to the meta-regression); whether weaker designs are more common where low-income economies are sampled; whether study quality goes with the reported direction; which enabling conditions authors test and in which direction (136 heterogeneity entries classified by one model reader, with a second reader on a random 20% sample); the thresholds stated; an evidence-gap map by outcome family and income group; and a comparison of the AI and historical-wave streams. These analyses are exploratory, are not corrected for multiple testing, and should be read as hypotheses.

2.7 Macro-panel analysis for questions the review could not answer

Four questions (the shape of the effect by adoption level, effects by income group under one design, timing around technology arrival, and differences inside low-income economies) cannot be answered from the included studies. We therefore specified, before downloading the data, a small analysis of public World Bank World Development Indicators for 1995 to 2019 (`analysis/spec.md`). In five-year periods, annual growth of GDP per capita was regressed on the change in mobile subscriptions per 100 people over the same period (A1) or the previous period (A2), with initial income, initial secondary enrolment, change in trade openness and period effects, by current World Bank income group, with standard errors clustered by country. A3 splits the change by baseline adoption level, A6 adds enabling conditions inside low and lower-middle income economies, and A4 is a stacked event study around the year mobile subscriptions first reach 10 per 100. These are associations; we have no instrument. Cost-effectiveness cannot be examined with these data.

3 Results

3.1 Study selection

Table 1 gives the flow. Of 247 records sent to full-text assessment, 106 full texts were obtained and read, together with one review. After the second pass, 67 studies are included, 39 were excluded at full text (12 for too few economies, 5 for an ineligible exposure, 1 for an ineligible outcome, 21 for being conceptual or lacking their own estimation) and one was kept as context. Three studies (298, 878 and 1495) were moved from include to exclude after the second pass on eligibility grounds; this should be confirmed by a human reviewer. About 140 of the 247 records have not been read because no free full text was available.

Table 1: Study selection (PRISMA 2020 flow, counts from the project repository).

Stage	n
Records identified (OpenAlex 1,840; Crossref 1,285; NBER 579; before cross-source deduplication 3,704)	3,704
Duplicates removed across sources	92
Records after deduplication	3,612
Excluded by automated stage-1 rules	1,545
Read at title level / excluded at title level	2,067 / 1,611
Assessed at abstract level / excluded at abstract level	456 / 121
Held as duplicates, context, environmental-only, reviews, supplementary	88
Sent to full-text assessment	247
Full texts read (106 of the 247 plus one review)	107
Excluded at full text	39
Kept as context only	1
Included	67

3.2 Characteristics of included studies

The 67 included studies were published between 2003 and 2026, and 52 of them from 2020 onward. Fifty-four study ICT, mobile, internet or broadband; 11 study AI; 2 cover both. Most use country-year panels (48), with a few at industry-country level. In a rough classification of the main estimator, 25 use system or difference GMM, 24 use fixed-effects, random-effects or pooled OLS, 11 use ARDL, cointegration or heterogeneous-panel estimators and 7 use growth accounting, data envelopment or other methods. Thirty-six studies have no identification strategy beyond controls and fixed effects. Twenty list at least one low-income economy; studies that do not state their income coverage are counted as not listing one. Outcome families, with a study able to contribute to several, are: output and productivity 37, labour market 14, living standards beyond income 12, poverty and distribution 11, structural change 5, resource cost 4. Appendix A lists all studies.

3.3 Risk of bias

No study was rated at low risk of bias. The first-pass ratings were moderate for 8 studies, serious for 50 and critical for 9. In the second pass, the separate reader agreed with the overall rating for 55 of 64 studies, found six too harsh (268, 302, 323, 408, 707 and 1469) and three too lenient (57, 957 and 1031). We did not change any rating, and these differences should be settled by a human reviewer. The commonest weaknesses are adoption treated as exogenous, indices whose scope does not match the technology, and many specifications with no stated preferred one.

3.4 Accuracy of the extraction

Across 372 result entries in the 64 re-read records, 348 were confirmed, 19 corrected and 5 could not be verified; 13 of 133 heterogeneity entries were corrected. Of 203 regression estimates, 202 matched the printed tables and one needed a corrected income label. Corrections were mostly misread labels, counts, and claims that the source text does not make. These were applied or flagged in the repository. Because both passes used the same family of model reading the same machine-converted text, a human check of a sample remains necessary.

3.5 Output and productivity

Thirty-seven studies cover output or productivity (4 on AI). Most report a positive association, but the sign is not uniform: current and lagged mobile effects have opposite signs in one 41-economy GMM study (study 10); ten high-income countries show no significant effect of any ICT indicator (study 1448); and in 27 EU states the digitalisation coefficient loses significance once economic freedom is added (study 988). Interaction terms are the most repeated finding: ICT pays more where human capital is higher in two Asian GMM studies (302 and 323, which share authors, panel and index), a developing-country productivity study, and in Africa, where the effect of the digital economy is negative below a human-capital index of about 2.35 and the average country is below it. Financial development, trade openness, governance, electricity and the diffusion level of ICT also appear as moderators. These are conditional associations from single-equation panels and do not show that raising a moderator would change the return. A non-parametric study of 24 OECD economies finds that ICT capital change has no contemporaneous effect on the production frontier and a positive effect five years later.

3.6 Structural change, labour markets, distribution, living standards and resource cost

Structural change (5 studies). Internet penetration is associated with faster structural change in a 51-economy panel, more so in low and lower-middle income and agrarian economies. A study of about 100 developing economies finds that ICT lowers export concentration, with a larger effect in the low and lower-middle income group (-0.155 on the concentration index) than in the high and upper-middle group (-0.020), but no formal test of the difference.

Labour market (14 studies, 5 on AI). In advanced economies ICT is linked to routine-task decline and polarisation, and capital-skill complementarity raises skill premia. No included study measures routinisation in low-income economies. Third-generation mobile coverage raises women's labour-force participation but not men's in 14 low- and middle-income countries, in one of the few studies that uses an instrument. AI exposure is positively associated with employment growth, concentrated in high-education occupations and younger workers, in European samples; in Southeast Asia the pattern differs by country, though the country-level estimates could not be checked in the source figure.

Poverty and distribution (11 studies, 2 on AI). Results split. A threshold model for 45 developing countries finds that ICT raises the Gini coefficient below a digital-maturity score of 0.48 and lowers it above. In 48 African economies ICT is positively associated with a composite inclusive-growth index. Robots raise carbon inequality between countries, less so in open economies. The thresholds rest on single studies.

Living standards (12 studies, 3 on AI). Mobile subscriptions are positively associated with the Human Development Index in high-income and lower-middle-income groups and not in low-income countries, in a large panel. In low and lower-middle income countries ICT narrows gender education gaps up to about 100 mobile subscriptions per 100 people, after which the gap widens. Internet effects on financial inclusion rise with governance quality in sub-Saharan Africa.

Resource cost (4 studies, 2 on AI). Digitisation is associated with higher carbon productivity, and the effect falls with income (0.085 for high-income, 0.058 for middle-income, 0.041 for low-income countries), while a digital-economy index lowers green total factor productivity in Belt and Road economies. Four studies are too few to generalise.

3.7 Do gains differ by income level?

Twenty-six studies compare income groups or regions, formally or descriptively (Table 2). Eight find larger effects in richer or OECD economies, six in poorer or less productive ones, and three find no clear difference; one finds a negative effect in all groups. Several of the first group have no test of the difference, and the studies finding larger effects in poorer economies include one single cross-section rated critical. The remaining studies compare groups without a directional finding, and study 268 appears in both of the first two groups because its direct and moderating effects point in different directions. We therefore treat the evidence as inconclusive. Studies of richer economies mostly use OLS or fixed effects, test the difference informally, and rarely include low-income economies, so the lean toward larger measured gains in richer samples may reflect study design.

Table 2: Studies that compare income groups or regions (study IDs in Appendix A).

Finding	Studies	Notes
Larger in richer or OECD	88, 138, 162, 268 (direct effect), 276, 535, 750, 957	Several lack a test of the difference; 535 rated critical and its figures look illustrative; 957 compares OECD with MENA
Larger in poorer or less productive	268 (informality moderation), 341, 407, 707, 952, 1192	407 compares Central and Eastern with Western Europe; 1192 is a single cross-section rated critical; 952 flips sign between pooled and subgroup models
No clear difference	29, 320, 879	879 compares low with lower-middle income only
Negative in all groups	245	Green total factor productivity outcome

3.8 Evidence on AI

The 13 studies on AI (11 AI only, 2 both streams) are mostly short panels of 3 to 78 economies, typically starting in 2012 or later, with exposure measured by occupational exposure indices, national readiness indices, patent-based measures or industrial robot density. None measures realised use of generative AI at country level. Where AI studies show an effect, it tends to be larger in richer economies and in high-skill occupations. The evidence from low-income economies is one productivity study of 63 low and lower-middle income economies and two studies rated critical. AI cannot be compared with earlier waves on equal terms.

3.9 Pilot meta-regression: ICT-type adoption and output

Of 206 estimates from 26 papers, 103 from 18 papers were usable. For the 16 papers with a convertible headline estimate, the random-effects pooled partial correlation is 0.189 (95% CI 0.104 to 0.274), with $\tau^2 = 0.027$, $I^2 = 93\%$ and a 95% prediction interval from -0.141 to 0.519 . A partial correlation of 0.19 corresponds to about 3.6% of residual variance. Of the headline estimates, 88% are positive and 75% are positive and statistically significant.

Table 3 shows that GMM estimates are on average 0.16 lower in partial correlation than estimates from other methods. This is consistent with reverse causation inflating simpler estimates, though it cannot prove it. The low- or middle-income indicator is not significant, but only 5 papers (11 estimates) come from such samples, against 16 papers and 92 estimates for high-income or mixed samples (mean partial correlation 0.031 against 0.123, unweighted). The

Table 3: Moderator meta-regression of partial correlations (103 estimates, 18 papers, standard errors clustered by paper).

Moderator	Coefficient	Clustered SE	p
Constant	0.087	0.056	0.12
Low- or middle-income sample	−0.038	0.045	0.40
System or difference GMM	−0.163	0.048	< 0.01
Mobile (relative to general ICT)	0.053	0.052	0.31
Internet or broadband (relative to general ICT)	0.135	0.079	0.09

FAT-PET slope on the standard error is 1.91 ($p = 0.06$) with a bias-corrected intercept of 0.005 ($p = 0.83$), and the PEESE intercept is 0.064 ($p = 0.04$), so the pooled value may be substantially inflated by selective reporting (Figure 1). The pooled estimate is stable to the degrees-of-freedom assumption (0.186 to 0.193 for 0, 5 and 20 regressors).

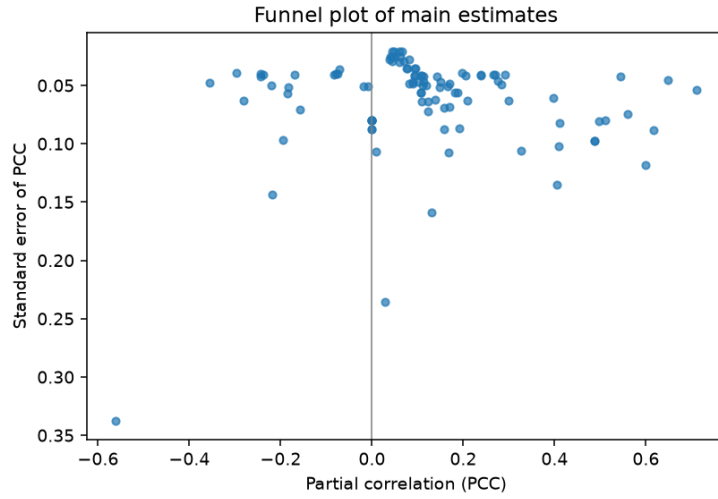


Figure 1: Funnel plot of main estimates (partial correlation against its standard error).

3.10 Exploratory analyses of additional questions

Measurement, outcome form and period. Adding single variables to the meta-regression, composite-index exposures show a partial correlation 0.07 higher than single indicators (SE 0.11, $p = 0.50$) and growth-rate outcomes 0.04 higher than level outcomes (SE 0.10, $p = 0.67$); neither is distinguishable from zero, and both rest on between-paper contrasts because few papers contribute both kinds. Later samples show lower partial correlations (-0.014 per year of sample midyear, SE 0.009, $p = 0.11$), which would be about -0.14 per decade. This is not significant, is partly confounded with the shift to GMM in recent studies, and is a hypothesis about diminishing returns, not a finding.

Design and income coverage. Studies that list a low-income economy were not more likely to lack an identification strategy (8 of 20, 40%, against 28 of 47, 60%; Fisher $p = 0.18$) or to be rated at higher risk of bias (moderate risk: 2 of 20 against 6 of 47, $p = 1.00$). We therefore find no evidence that weaker designs cluster where low-income economies are sampled, although the comparison has little power. The test of whether study quality goes with a positive direction for output could not be run: only one of the 37 output studies was rated at moderate risk of bias.

Enabling conditions. Of the 136 heterogeneity entries, 105 are moderation tests. Human capital is tested in 14 studies (amplifies the effect in 8, dampens it in 3, mixed in 4), income group in 18 (richer or higher-income group larger in 7, smaller in 3, mixed in 3, no difference in 4, unclear in 2), infrastructure or connectivity in 8 (5 amplify), institutions or governance in 8 (3 amplify, 2 dampen, 4 mixed), trade openness in 6 (3 amplify, 2 dampen), financial development in 5 and sector structure in 8 (6 mixed). A second reader agreed with the moderator type in 86% of a 28-entry sample (kappa 0.83) but with the direction in only 71% (kappa 0.64), so the direction counts are indicative only. Amplification is the commonest finding for human capital and infrastructure, which is consistent with, but does not show, conditional returns.

Thresholds. The studies state 21 thresholds in 14 studies, for the human-capital index, digital maturity, internet penetration, trade openness, private credit and others. They are in different units, rest on one study each and cannot be pooled. Where two studies address the same variable they do not give comparable values.

Evidence gaps. Table 4 shows that low-income economies are listed in 11 of 37 output studies, 3 of 14 labour-market studies, 2 of 11 distribution studies and 2 of 4 resource-cost studies. High-income economies are listed in 22 of 37 output studies and 11 of 14 labour-market studies. The historical-wave and AI streams look alike on income-group comparison (39% against 38%) but differ on risk of bias (critical in 9% against 31%) and first sample year (median 2000 against 2012); the AI stream lists a low-income economy in 2 of 13 studies against 18 of 54 (Fisher $p = 0.31$).

Table 4: Evidence-gap map: studies per outcome family by income groups listed in the sample (a study counts in each group it lists).

Outcome family	Low	Lower-middle	Upper-middle	High	Not reported	Studies
Output and productivity	11	11	11	22	9	37
Structural change	3	3	3	3	1	5
Labour market	3	5	6	11	2	14
Poverty and distribution	2	1	3	7	4	11
Living standards	4	4	4	6	4	12
Resource cost	2	1	2	4	0	4

3.11 Macro-panel analysis

Table 5 shows that a 10-subscription increase in mobile adoption per 100 people goes with higher annual growth in all four income groups over the same five-year period. The association is largest in lower-middle income economies (0.47 percentage points, 95% CI 0.16 to 0.77) and about 0.16 in upper-middle and high-income economies; the low-income estimate (0.36) is imprecise (CI -0.05 to 0.76, 20 economies). With country fixed effects, the lower-middle estimate remains (0.36, $p = 0.03$), and extending the sample to 2023 gives the same pattern. Across baseline adoption levels, the association is largest where adoption was below 10 per 100 (0.41) and smaller at 40 to 80 (0.18); it is not monotone, since it is 0.27 above 80 per 100. Inside low and lower-middle income economies, none of initial secondary enrolment, electricity access, private credit or initial income significantly changed the association (all $p > 0.19$, wide intervals).

The predetermined specification, which uses the previous period’s change, does not reproduce this pattern: the association is positive only in lower-middle income economies (0.26, $p = 0.02$), near zero in upper-middle, and negative in low-income (-0.38 , $p = 0.13$) and high-income (-0.16 , $p = 0.01$) economies, and negative in every income group for internet use. The contemporaneous

associations therefore partly reflect growth driving adoption or shared shocks. The event study around mobile takeoff could not be interpreted: nearly every economy (189 of 190 below the threshold in 1995) crosses it in the window, there are few clean controls (7 in the low and lower-middle group), and that group shows a pre-trend of 12 log points before takeoff. Results for fixed broadband rest on extrapolation from very few observations and are not interpreted.

Table 5: Mobile adoption and annual GDP-per-capita growth by income group: change in annual growth (percentage points) per +10 subscriptions per 100 people, five-year periods 1995–2019 (95% CI; standard errors clustered by country).

Income group	Economies	Same period (A1)	Previous period (A2)
Low	20	0.36 [−0.05, 0.76]	−0.38 [−0.86, 0.11]
Lower-middle	42	0.47 [0.16, 0.77]	0.26 [0.03, 0.49]
Upper-middle	51	0.16 [0.04, 0.28]	0.01 [−0.15, 0.16]
High	64	0.16 [0.05, 0.26]	−0.16 [−0.28, −0.03]

4 Discussion

Three points stand out. First, the central estimate of the association between ICT-type adoption and output is positive but modest and very heterogeneous across studies, and it shrinks when we allow for small-study bias and for estimators that deal with reverse causation. Second, returns look conditional on human capital, financial depth, governance and infrastructure. This agrees with the long-run diffusion evidence that intensity of use, rather than arrival, drives divergence [3], although the studies we synthesise cannot show that changing a moderator would change the return. Third, the question of whether poorer or richer economies gain more cannot be answered from this evidence. The comparisons that exist are few, mostly informal and often from weak designs, and low-income economies are rarely sampled.

The panel analysis adds one cautious observation. Under a single design, the association between mobile adoption and growth is not smaller in poorer economies: it is largest in lower-middle income economies. This runs against the lean in the reviewed studies toward larger measured gains in richer samples, but the predetermined version does not reproduce it, so we do not take it as evidence that poorer economies gain more.

For policy, the safest reading is that digital investment is unlikely to pay off equally everywhere and that complementary capabilities matter. The review does not support statements of the form that a given income group should expect a given payoff. For research, the main need is identification for developing and low-income economies, with instruments or timing designs, and for AI, measures of realised use rather than readiness.

5 Limitations

- **Selection.** Only studies with a free or supplied full text were read: about 140 of the 247 records sent to full-text assessment are unread, and paywalled studies are over-represented among them. Records from four venues the author judged unreliable were skipped. Search covered OpenAlex, Crossref and NBER only; Scopus, Web of Science and EconLit were not run.

- **Single reviewer.** Screening used one assisted reviewer, so no agreement statistic exists. Extraction was by machine with a second machine pass; no human has verified the data.
- **Registration.** The protocol was not registered before the search.
- **Design quality.** No study has low risk of bias; most treat adoption as exogenous. We report associations, not causal effects. First-pass risk-of-bias ratings are used; nine ratings would change under the second pass.
- **Counting.** Direction counts are not weighted by size or quality and are not effect sizes. Study-level income comparisons mix World Bank income groups, regions and developed-developing splits.
- **Meta-regression.** Eighteen papers; the clustered standard errors are optimistic. Partial correlations rely on an assumed degrees of freedom. Composite indices, levels and growth outcomes, and conditional main effects are mixed in one analysis despite exclusions. Two Asian studies (302 and 323) share authors, panel and index and are not independent. The sample is not a census, and the low- and middle-income group is small.
- **Language and period.** English only, 1995 onward.
- **Macro-panel analysis.** The World Bank panel analysis gives conditional associations only. Income groups use the current classification; the planned check with initial income was not run; CO2 emissions could not be retrieved; and the event-study and broadband results are not interpretable. Roughly thirty estimates were reported without correction for multiple testing.

6 Conclusion

Across 67 studies, ICT-type adoption is usually associated with higher output, but the association is modest, strongly heterogeneous, shrinks under bias corrections and estimators that address reverse causation, and is conditional on human capital, financial depth, governance and infrastructure. Whether rich or poor economies gain more is not established, and the evidence on AI is too thin to compare with earlier waves. Credible identification for low- and middle-income economies and realised-use measures for AI are the main gaps.

Declarations

Data and code availability. The protocol, search and screening logs, extraction database, second-pass verification files, meta-regression inputs and code are in the project repository: <https://github.com/vinhqdang/Journal-of-Economy-and-Technology>. Full-text PDFs are not shared for copyright reasons.

Ethics. Not applicable; the review uses published studies only.

Author contributions (CRediT). [author to complete]

Conflict of interest. [author to complete]

Funding. No external funding was declared in the protocol. [author to confirm]

Use of AI tools. [author to complete the disclosure required by the journal’s policy on generative AI. Screening and extraction used automated assistance; the wording and the level of detail are the author’s decision.]

Registration. [OSF registration pending]

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A Included studies

Table 6: Studies included in the synthesis (n = 67). Authors are not listed because the extraction database does not store them; the DOI identifies each record. n.r. = not reported. RoB = overall risk of bias in the first-pass rating (moderate, serious, critical); no study was rated low.

ID	Year	Title	Technology	Econ.	RoB	DOI
10	2020	Financial inclusion, information and communication technology diffusion, and economic growth: a panel data analysis	ICT, Mobile, Internet	41	serious	https://doi.org/10.1080/02681102.2020.1734770
29	2019	The Internet of Things and economic growth in a panel of countries	ICT, Mobile, Internet	82	serious	https://doi.org/10.1080/10438599.2019.1695941
45	2024	Financial inclusion through digitalization and economic growth in Asia-Pacific countries	ICT, Mobile, Internet	30	serious	https://doi.org/10.1016/j.irfa.2024.103596
57	2020	Nexus between energy consumption, information and communications technology, and economic growth: An enquiry into emerging Asian countries	ICT, Mobile, Internet	10	serious	https://doi.org/10.1002/pa.2172
68	2022	Enhancing human development in developing regions: Do ICT and transport infrastructure matter?	ICT, Mobile, Internet	79	serious	https://doi.org/10.1016/j.techfore.2022.121725

ID	Year	Title	Technology	Econ.	RoB	DOI
71	2022	The ICT, financial development, energy consumption and economic growth nexus in MENA countries: dynamic panel CS-ARDL evidence	ICT, Mobile, Internet	11	serious	https://doi.org/10.1080/00036846.2022.2096861
88	2005	Does the digital divide matter? The role of information and communication technology in cross-country level and growth estimates	ICT, Mobile, Internet	94	serious	https://doi.org/10.1080/1043859042000304043
108	2024	New technologies and jobs in Europe	AI	16	moderate	https://doi.org/10.1093/epolic/eiae058
126	2023	Job creation and destruction in the digital age: Assessing heterogeneous effects across European Union countries	ICT	27	moderate	https://doi.org/10.1016/j.econmod.2023.106405
138	2023	Empirical Analysis of Inclusive Growth, Information and Communication Technology Adoption, and Institutional Quality	ICT, Mobile, Internet	193	serious	https://doi.org/10.3390/economies11040124
140	2015	What are the drivers of total factor productivity in the European Union?	ICT	17	serious	https://doi.org/10.1080/10438599.2015.1067007
162	2024	Impact of digitization on carbon productivity: an empirical analysis of 136 countries	Mobile, Internet	136	serious	https://doi.org/10.1038/s41598-024-55848-2
174	2025	Artificial intelligence and global carbon inequality: Addressing the challenges and opportunities for SDG 10, SDG 12, and SDG 13	AI	67	serious	https://doi.org/10.1016/j.gsf.2025.102072
217	2021	Does digital financial inclusion moderate or exacerbate output volatility?	Mobile	40	serious	https://doi.org/10.1080/13504851.2021.1963400
239	2018	What drives labor market polarization in advanced countries? The role of China and technology	ICT	19	serious	https://doi.org/10.1093/icc/dty063
245	2023	The impact of the digital economy on green total factor productivity in Belt and Road countries: the mediating role of energy transition	ICT, Internet	40	serious	https://doi.org/10.3389/fenvs.2023.1213961
268	2023	Informality and aggregate labor productivity growth: Does ICT moderate the relationship?	ICT, Internet	118	serious	https://doi.org/10.1016/j.telpol.2023.102681
269	2024	Economic growth and the proliferation of ICT infrastructures: which causes the other?	ICT, Mobile, Internet	41	serious	https://doi.org/10.1080/1331677x.2024.2310100
276	2025	Does artificial intelligence help in improving human capital based educational development? Evidence from 29 countries	AI	29	serious	https://doi.org/10.1016/j.techsoc.2025.103004
278	2024	Towards ICT diffusion and trade liberalisation on inclusive growth in Sub-Saharan Africa	ICT, Mobile, Internet	48	serious	https://doi.org/10.1007/s10668-023-04355-x
285	2022	ICT and Economic Growth in EU: A macro Level Comparison of Estimated ICT Output Elasticities	ICT	27	moderate	https://doi.org/10.1080/1097198x.2022.2094182

ID	Year	Title	Technology	Econ.	RoB	DOI
302	2013	Does ICT Participate in Economic Convergence among Asian Countries: Evidence from Dynamic Panel Data Model	ICT, Mobile, Internet	24	serious	https://doi.org/10.12948/issn14531305/17.2.2013.01
320	2022	The Effect of Technology Adoption on Financial Inclusion: A Cross-country Panel Analysis between China and Nigeria	Mobile, Internet	20	serious	https://doi.org/10.24018/ejbmr.2022.7.2.1314
321	2024	Information and communication technology and labour productivity growth: a production-frontier approach	ICT	24	serious	https://doi.org/10.1007/s10479-024-05818-8
323	2014	Total Factor Productivity, Demographic Traits and ICT: Empirical Analysis for Asia	ICT, Mobile, Internet	24	serious	https://doi.org/10.12948/issn14531305/18.1.2014.01
341	2024	The impact of information and communication technologies on export diversification: Evidence from developing countries	ICT, Mobile, Internet	110	moderate	https://doi.org/10.1080/09638199.2024.2419406
358	2023	Digital Divide, Credit Financing and Financial Inclusion: Changing Patterns of Poverty, Employment and Income Inequalities	Mobile, Internet	148	serious	n.a.
392	2018	TECHNOLOGICAL SOURCES OF ECONOMIC GROWTH IN EUROPE AND THE U.S.	ICT	10	serious	https://doi.org/10.3846/20294913.2017.1280557
393	2013	The Determinants of Total Factor Productivity in the EU: Insights from Sectoral Data and Common Dynamic Processes	ICT	17	serious	n.a.
407	2015	The Role of ICT in the Productivity of Central and Eastern European Countries: Cross-country Comparison	ICT, Internet	21	serious	n.a.
408	2025	Effects of the digital economy on economic growth in Africa: the role of human capital	ICT, Mobile, Internet	41	serious	https://doi.org/10.1080/23322039.2025.2525485
422	2024	Icts And Labour Productivity Nexus In Developing Countries: Evidence From Panel Estimation Approach	ICT, Mobile, Internet	84	serious	https://doi.org/10.33736/ijbs.6868.2024
441	2024	Artificial Intelligence Readiness and Employment: A Global Panel Analysis	AI	78	critical	https://doi.org/10.24818/18423264/58.4.24.04
442	2025	From Digital Divide to Employment Equity: How Digitalization Affects Women’s Involvement in the Workforce in OECD Countries	Mobile, Internet	22	critical	https://doi.org/10.17583/generos.15672
535	2026	The Economic Value of Agentic AI: A Comparative Analysis of Its Impact on Growth and Business Productivity in Developed and Emerging Economies	AI	10	critical	https://doi.org/10.9734/ajrcos/2026/v19i5859

ID	Year	Title	Technology	Econ.	RoB	DOI
617	2003	Bridging the digital divide: the growth implications of e-commerce for small and developing states	Internet	205	critical	n.a.
707	2026	Digitalisation and structural change: Evidence from cross-country analysis	Mobile, Internet	51	serious	https://doi.org/10.1016/j.jdec.2026.03.003
750	2025	Artificial Intelligence and Labour Markets in Southeast Asia: An Empirical Examination	AI	1	serious	https://doi.org/10.46557/001c.132415
802	2015	Differences in Job De-Routinization in OECD Countries: Evidence from PIAAC	ICT	22	serious	n.a.
803	2020	The Race between Technological Progress and Female Advancement: Changes in Gender and Skill Premia in OECD Countries	ICT	11	moderate	n.a.
879	2026	AI capability and labor productivity in low- and middle-income countries: An absorptive capacity approach	AI, ICT, Mobile, Internet	63	serious	https://doi.org/10.1016/j.technovation.2026.103645
913	2026	Estimation of Firm Labour Productivity and Sales Growth from Artificial Intelligence in Sub-Saharan African Countries	AI	10	critical	https://doi.org/10.12688/f1000research.179912.1
952	2026	Twin Transition and Labor Informality in Asia-Pacific: Threshold Effects and Income-Group Heterogeneity	ICT, Internet	17	serious	https://doi.org/10.3390/economies14090398
957	2024	IMPACT OF ICT ON ECONOMIC GROWTH: CASE OF OECD AND MENA COUNTRIES	ICT, Mobile, Internet	32	serious	https://doi.org/10.59671/dga69
988	2023	IMPACT OF DIGITALISATION ON PRODUCTIVITY GROWTH IN EU MEMBER STATES	ICT	27	serious	https://doi.org/10.30525/2661-5169/2023-3-2
1013	2025	The Effect of Artificial Intelligence Investments on Economic Development: Panel Data Analysis	AI	8	serious	https://doi.org/10.17694/bajece.1802500
1025	2005	Testing the impact of ICT in developed countries during 1980-1995: distributional analysis in Solow's growth accounting framework	ICT	15	serious	n.a.
1026	2026	The nonlinear link between AI and energy poverty: Empirical evidence from global data	AI	64	serious	https://doi.org/10.1016/j.jclepro.2026.148590
1031	2025	Exploring the Moderating impact of ICT Infrastructure on Trade Openness-Sectoral Growth Nexus in Sub-Saharan Africa (SSA)	ICT, Mobile, Internet	26	serious	https://doi.org/10.58709/niujhu.v10i3.2267
1192	2026	The Digital Maturity Paradox: The Divergent Impact of Network Readiness and FDI Across Development Tiers	ICT	121	critical	https://doi.org/10.3390/economies14080320

ID	Year	Title	Technology	Econ.	RoB	DOI
1372	2024	The relationship between inequality and poverty in developing countries: mitigating role of virtual social network and internet access in schools	Internet	57	serious	https://doi.org/10.1108/ijse-09-2023-0695
1419	2017	Mobile banking usage, quality of growth, inequality and poverty in developing countries	Mobile	93	serious	https://doi.org/10.1177/0266666917744006
1428	2021	IMPACT OF MOBILE PHONES AND INTERNET USE ON FINANCIAL INCLUSION: EMPIRICAL EVIDENCE FROM THE EU POST-COMMUNIST COUNTRIES	Mobile, Internet	11	serious	https://doi.org/10.3846/tede.2021.14508
1438	2021	Impact of Economic Integration and Information and Communication Technology on Economic Growth for European Union: Dynamic Panel GMM Approach	ICT, Mobile, Internet	27	serious	https://doi.org/10.5296/ber.v11i4.19064
1448	2023	The effect of information and communication technology on economic growth high-income countries	ICT, Mobile, Internet	10	serious	https://doi.org/10.55493/5002.v13i9.4824
1469	2026	Reducing Gender Inequality in Education in Developing Countries: Do Information and Communication Technologies Matter?	ICT, Mobile, Internet	64	serious	https://doi.org/10.1002/sd.71201
1494	2026	Digital transformation and income disparities: A threshold analysis in developing countries	ICT, Mobile, Internet	45	serious	https://doi.org/10.1556/032.2025.00183
1526	2022	Artificial Intelligence and Employment: New Cross-Country Evidence	AI	23	moderate	https://doi.org/10.3389/frai.2022.832736
1564	2025	The Impact of Artificial Intelligence on Employment: A Panel Data Analysis for Selected Countries	AI	29	serious	https://doi.org/10.30784/epfad.1621455
1596	2026	Impact of Generative AI Adoption on Micro and Small Enterprises in Emerging Economies	Other	3	critical	https://doi.org/10.64060/ijpssdv2i12
1645	2025	Cryptocurrency Adoption Patterns and Financial Inclusion Outcomes in Emerging Market Economies	Mobile, Internet, Crypto	15	critical	https://doi.org/10.71086/iajir/v12i4/iajir1244
1652	2025	Crypto-driven growth: A comparative study of Bitcoin and Ethereum on economic growth for multi-country analysis	Crypto	14	critical	https://doi.org/10.32609/j.ruje.11.164511
1670	2019	FINANCIAL INCLUSION, MOBILE PHONE DIFFUSION, AND ECONOMIC GROWTH; EVIDENCE FROM AFRICA	Mobile	32	serious	https://doi.org/10.32479/ijefi.8256
1679	2022	The Employment Effects of Mobile Internet in Developing Countries	Mobile, Internet	14	moderate	n.a.

ID	Year	Title	Technology	Econ.	RoB	DOI
1713	2012	ICT capital and labour productivity growth: A non-parametric analysis of 14 OECD countries	ICT	14	serious	https://doi.org/10.1016/j.telpol.2011.12.012
1759	2022	ICT capital–skill complementarity and wage inequality: Evidence from OECD countries	ICT	14	moderate	https://doi.org/10.1016/j.labeco.2022.102151
1798	2022	The role of governance in the effect of the internet on financial inclusion in sub-Saharan Africa	Internet	42	serious	https://doi.org/10.2139/ssrn.4315622

B Protocol amendments

1. (2 Oct 2026) The OpenAlex query was written with explicit phrase variants and restricted to the economics field, because an unrestricted query returned about 7,600 records, too many for one screener.
2. (2 Oct) Four holding categories were added at the abstract stage: context, environmental outcome only, reviews for backward citation search, and supplementary micro evidence.
3. (2 Oct) arXiv was attempted but not completed. Scopus, Web of Science and EconLit were not searched.
4. (2 Oct) Automated stage-1 rules were added before manual reading, with a single assisted screener and a verification sample of 90 decisions.
5. (2 Oct) Included records were ranked in two tiers to decide which paywalled studies to request first.
6. (2 Oct) Records with no free full text and not on the short request list were kept as an abstract-level evidence map, excluded from pooled estimates and from risk-of-bias ratings.
7. (3 Oct) Studies whose only available venue the author judged unreliable were not read and not used.
8. (3 Oct) Three studies (298, 878 and 1495) were moved from include to exclude after a second-pass check. Five regression estimates were excluded from the meta-regression after the same check.